

AASD 4016

Full Stack Data Science Systems

Applied AI Solutions Developer Program



Module 1

Scripting Notebooks

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Agenda



- Python Notebooks
- Python Scripts
- IDE
- Scripting process
- Examples

Python Notebooks

.ipynb



IPython Notebook

Import the necessary library

```
In [1]: import PyPDF2
```

PDF file

```
In [2]: filename = "RBC-10_questions_for_first_home-buyers.pdf"
```

Read the PDF file

```
In [3]: pdfFileObj = open(filename, 'rb')
```

Use PdfFileReader

```
In [4]: pdfReader = PyPDF2.PdfFileReader(pdfFileObj)
```

```
PdfReadWarning: Xref table not zero-indexed. ID numbers for objects will be corrected. [pdf.py:1736]
```

Get the number of pages

```
In [5]: num_pages = pdfReader.numPages
```

Initialize count and text

```
In [6]: count = 0  
text = ""
```

Python Scripts

.py



Python Script

```
def main():  
    filename = "inputs/00_tts_utterances.csv"  
    df = pd.read_csv(filename)  
    df['AudioFile'] = df['Reference'].apply(tts)  
  
    df.to_csv("inputs/tts.csv")  
    upload_blob('phonetic-discovery',  
               f"inputs/tts.csv",  
               "tts.csv")  
    # tts(text)  
  
if __name__ == "__main__":  
    main()
```

IDE

VSCode

Sublime Text

Atom

Vim

Spyder

Integrated Development Environment

VSCode

Atom

Sublime Text

Spyder

Vim

Scripting Process

.ipynb to .py



Scripting Process

IPython Notebooks (.ipynb) to be converted into Python Scripts (.py)

Pros

- Works well in Production environment
- Can be run from terminal
- Can be made as an API to be served to consumers

Process

- See what is the problem that the notebook trying to solve
- Find the inputs and outputs of the entire problem
- Find the steps to solve that problem
- Create an **app.py** file
- Place the notebook cells in the app.py file
- Clean the app.py file
 - Remove display of variables
- Find a way to modularize the above found steps into smaller functions
- Place the functions in different python scripts (if possible and makes sense)
- Import those functions in the app.py
- Once app.py is clean and finished, run it in the terminal
 - `python app.py`

Examples

01_Scripting_Notebook.ipynb to 01_app.py

02_Scripting_Notebook.ipynb to 02_app.py

03_Scripting_Notebook.ipynb to 03_app.py

Example 1 Scripting Process

- 01_Scripting_Notebook.ipynb
 - Look at the example notebook
 - **PDF file to Text file Conversion**
- 01_app.py
 - Blank python script (to be filled from the notebook)

Example 2 Scripting Process

- 02_Scripting_Notebook.ipynb
 - Look at the example notebook
- 02_app.py
 - Blank python script to be filled from the notebook

Example 3 Scripting Process

- 03_Scripting_Notebook.ipynb
 - Look at the example notebook
- 03_app.py
 - Blank python script to be filled from the notebook